

* Cartesian Product

For sets A, B the Cartesian product or Cross product of A and B is denoted by $A \times B$ and is defined as
 $A \times B = \{(a, b) : a \in A, b \in B\}$.

1. If $A = \{1, 2, 3\}$ $B = \{3, 4, 5\}$ $C = \{4, 5, 6\}$, then find (i) $A \times (B \cap C)$
 (ii) $(A \times B) \cap (A \times C)$ (iii) $A \times (B \cup C)$ (iv) $(A \times B) \cup (A \times C)$

Sol: (i) $A \times (B \cap C) = \{(1, 4) (1, 5) (2, 4) (2, 5) (3, 4) (3, 5)\}$

(ii) $A \times B = \{(1, 3) (1, 4) (1, 5) (2, 3) (2, 4) (2, 5) (3, 3) (3, 4) (3, 5)\}$

$A \times C = \{(1, 4) (1, 5) (1, 6) (2, 4) (2, 5) (2, 6) (3, 4) (3, 5) (3, 6)\}$

$\Rightarrow (A \times B) \cap (A \times C) = \{(1, 4) (1, 5) (2, 4) (2, 5) (3, 4) (3, 5)\}$

(iii) $A \times (B \cup C) = \{(1, 3) (1, 4) (1, 5) (1, 6) (2, 3) (2, 4) (2, 5) (2, 6) (3, 3) (3, 4) (3, 5) (3, 6)\}$

(iv) $(A \times B) \cup (A \times C) = \{(1, 3) (1, 4) (1, 5) (1, 6) (2, 3) (2, 4) (2, 5) (2, 6) (3, 3) (3, 4) (3, 5) (3, 6)\}$



(i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

2. If $A = \{1, 2, 3, 4\}$ $B = \{2, 5\}$ $C = \{3, 4, 7\}$, then find (i) $A \times B$
 (ii) $B \times A$ (iii) $(A \cup B) \times C$ (iv) $(A \times C) \cup (B \times C)$ (v) $(A \cap B) \times C$
 (vi) $(A \times C) \cap (B \times C)$

Sol: (i) $A \times B = \{(1, 2) (1, 5) (2, 2) (2, 5) (3, 2) (3, 5) (4, 2) (4, 5)\}$

(ii) $B \times A = \{(2, 1) (2, 2) (2, 3) (2, 4) (5, 1) (5, 2) (5, 3) (5, 4)\}$

(iii) $(A \cup B) \times C = \{(1, 3) (1, 4) (1, 7) (2, 3) (2, 4) (2, 7) (3, 3) (3, 4) (3, 7) (4, 3) (4, 4) (4, 7) (5, 3) (5, 4) (5, 7)\}$

(iv) $A \times C = \{(1, 3) (1, 4) (1, 7) (2, 3) (2, 4) (2, 7) (3, 3) (3, 4) (3, 7) (4, 3) (4, 4) (4, 7)\}$

$B \times C = \{(2, 3) (2, 4) (2, 7) (5, 3) (5, 4) (5, 7)\}$

$$(A \times C) \cup (B \times C) = \{(1, 3)(1, 4)(1, 7)(2, 3)(2, 4)(2, 7)(3, 3)(3, 4)(3, 7)(4, 3)(4, 4)(4, 7)(5, 3)(5, 4)(5, 7)\}$$

$$(v) (A \cap B) \times C = \{(2, 3)(2, 4)(2, 7)\}$$

$$(vi) (A \times C) \cap (B \times C) = \{(2, 3)(2, 4)(2, 7)\}$$



$$(A \cup B) \times C = (A \times C) \cup (B \times C)$$

$$(A \cap B) \times C = (A \times C) \cap (B \times C)$$



Relation

For sets A, B any subset of $A \times B$ is called a relation from A to B . Any subset of $A \times A$ is called a relation on A .



For finite sets A, B with $|A| = m$ and $|B| = n$, there are 2^{mn} relations from A to B , including empty relation as well as the relation $A \times B$ itself

1. If $A = \{a, b, c, d\}$ and $B = \{w, x, y\}$ (i) Give examples of three non-empty relations from A to B (ii) Any three examples of non-empty relations on A .

2. If $A = \{1, 2, 3\}$ and $B = \{2, 4, 5\}$, give examples of (i) three non-empty relations from A to B (ii) three non-empty relations on A .

The collection of all subsets of $A \times B$ is called power set of $A \times B$ and is denoted by $P(A \times B)$. Also $|P(A \times B)| = 2^{mn}$.

The number of relations from A to B that contain the pairs $(a_1, b_1)(a_2, b_2) \dots (a_k, b_k) = 2^{mn-k}$.

The number of relations from A to B that contains exactly ' k ' Ordered pairs $= {}^nC_k$ where $n = |A \times B|$.

1. Let $A = \{a, b, c, d, e, f, g, h\}$ and $B = \{1, 2, 3, 4, 5\}$. How many elements are there in $P(A \times B)$ 2.

Sol: Given $|A| = 8$, $|B| = 5 \Rightarrow |A \times B| = 8 \times 5 = 40$.

The number of Subset of $A \times B$ / number of relations from A to $B = 2^{40}$
ie $|P(A \times B)| = 2^{40}$.

2. Show that $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Pf: Let $(a, b) \in A \times (B \cap C) \Rightarrow a \in A$ & $b \in B \cap C$
 $\Rightarrow a \in A$ & $b \in B$ and $b \in C$
 $\Rightarrow (a \in A \text{ & } b \in B) \text{ and } (a \in A \text{ & } b \in C)$
 $\Rightarrow (a, b) \in A \times B \text{ and } (a, b) \in A \times C$
 $\Rightarrow (a, b) \in (A \times B) \cap (A \times C)$

ie $A \times (B \cap C) \subseteq (A \times B) \cap (A \times C)$

In the Similar way we can p.T $(A \times B) \cap (A \times C) \subseteq A \times (B \cap C)$

Hence $A \times (B \cap C) = (A \times B) \cap (A \times C)$

3. Show that $(A \cap B) \times C = (A \times C) \cap (B \times C)$

Pf: Let $(a, b) \in (A \cap B) \times C \Rightarrow a \in A \cap B$ & $b \in C$
 $\Rightarrow a \in A$ and $a \in B$ & $b \in C$
 $\Rightarrow (a \in A \text{ & } b \in C) \text{ and } (a \in B \text{ & } b \in C)$
 $\Rightarrow (a, b) \in A \times C \text{ and } (a, b) \in B \times C$
 $\Rightarrow (a, b) \in (A \times C) \cap (B \times C)$

ie $(A \cap B) \times C \subseteq (A \times C) \cap (B \times C)$

In the Similar way we can p.T $(A \times C) \cap (B \times C) \subseteq (A \cap B) \times C$

Hence $(A \cap B) \times C = (A \times C) \cap (B \times C)$

4. Show that $A \times (B \cup C) = (A \times B) \cup (A \times C)$

Pf: Let $(a, b) \in A \times (B \cup C) \Rightarrow a \in A$ and $b \in B \cup C$
 $\Rightarrow a \in A$ and $b \in B$ or $b \in C$

8. Let A, B be sets with $|B|=3$. If there are 4096 relations from A to B what is $|A|$.

Sol: Given $2^{mn} = 4096$, $n=3$

ie $2^{3m} = 4096$

ie $2^{3m} = 2^{12} \Rightarrow 3m=12$
 $\Rightarrow m=4. \Rightarrow |A|=4.$

9. If $A = \{1, 2, 3\}$ and $B = \{2, 4, 5\}$. Determine the following
 (i) $|A \times B|$ (ii) The number of relations from A to B
 (iii) The number of relations on A . (iv) The number of relations from A to B that contain $(1, 2)$ and $(1, 5)$ (v) The number of relations from A to B that contains exactly 5 ordered pairs (vi) The number of relations on A that contain atleast seven elements.

Sol: (i) $|A \times B| = |A| \cdot |B| = 3 \times 3 = 9.$

(ii) No. of relations from A to $B = 2^9$

(iii) No. of relations on $A = 2^9$

(iv) No. of relations from A to B that contain $(1, 2) \& (1, 5) = 2^{9-2} = 2^7 = 128$

(v) No. of relations from A to B that contains exactly 5 ordered pairs $= {}^9C_5 = 126$

(vi) No. of relations on A that contain atleast seven elements $= {}^9C_7 + {}^9C_8 + {}^9C_9$



Functions

For non-empty sets A, B a function or mapping f from A to B denoted by $f: A \rightarrow B$, is a relation from A to B in which every element of A appears exactly once as the first component of an ordered pair in the relation.

For the function $f: A \rightarrow B$, A is called the domain of f and B the codomain of f . The subset of B consisting of those elements that appears as second components in the ordered pair of f is called the range of f and is denoted by $f(A)$ because it is the set of images under f .

* Let A, B two non-empty sets with $|A|=m$ and $|B|=n$, then there are ' n^m ' functions from A to B .

* One-One function

A function $f: A \rightarrow B$ is called one to one or injective, if each element of B appears atmost once in the image of an element of A .

* If $|A|=m$ and $|B|=n$, then number of one to one functions from A to B is $P(n, m) = \frac{n!}{(n-m)!}$ ($n > m$)

* For arbitrary sets A, B $f: A \rightarrow B$ is one to one iff for all $a_1, a_2 \in A$, $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$.

1. If $A = \{1, 2, 3, 4, 5\}$ and there are 6720 injective functions $f: A \rightarrow B$. What is $|B|$.

Sol: Given $|A|=m=5$ and $P(n, m) = 6720$.

$$\Rightarrow \frac{n!}{(n-m)!} = 6720$$

$$\Rightarrow \frac{n!}{(n-5)!} = 6720$$

$$\Rightarrow |B|=n=8.$$

2. Let $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$

(i) List five functions from A to B (ii) How many functions $f: A \rightarrow B$ are there

(iii) How many functions $f: A \rightarrow B$ are one to one.

(iv) How many functions $g: B \rightarrow A$ are there. (v) How many functions

$f: B \rightarrow A$ are one to one. (vi) How many functions $f: A \rightarrow B$

Satisfy $f(1) = x$ (vii) How many functions $f: A \rightarrow B$ Satisfy $f(1) = f(2) = x$

(viii) How many functions $f: A \rightarrow B$ Satisfy $f(1) = x$ and $f(2) = y$.

Sol:

$$(i) \{ (1, x) (2, x) (3, x) (4, x) \}$$

$$\{ (1, y) (2, y) (3, y) (4, y) \}$$

$$\{ (1, z) (2, z) (3, z) (4, z) \}$$

$$\{ (1, x) (2, y) (3, z) (4, x) \} \quad \{ (1, x) (2, x) (3, y) (4, y) \}$$

(ii) Number of functions $f: A \rightarrow B = n^m = 3^4 = 81$.

(iii) Number of one-one functions $f: A \rightarrow B = 0$.

(iv) Number of functions $g: B \rightarrow A = m^n = 4^3 = 64$.

(v) Number of one-one functions $g: B \rightarrow A = P(4, 3) = \frac{4!}{(4-3)!} = 24$.

(vi) Number of functions $f: A \rightarrow B$ satisfy $f(1) = x$

Remaining no. of elements in the domain = 3

$\therefore$ No. of functions = $3^3 = 27$.

(vii) Number of functions $f: A \rightarrow B$ satisfy $f(1) = f(2) = x$

Remaining no. of elements in the domain = 2

$\therefore$ No. of functions = $3^2 = 9$.

(viii) Number of functions $f: A \rightarrow B$ satisfy $f(1) = x$ and $f(2) = y$

Remaining no. of elements in the domain = 2

$\therefore$ No. of functions = $3^2 = 9$.

3. Let $f: A \rightarrow B$ with $A_1, A_2 \subseteq A$. Then Prove that

(i) $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$

(ii) $f(A_1 \cap A_2) = f(A_1) \cap f(A_2)$ when f is one-one.

(iii) $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$

Sol:

(i) Let $b \in f(A_1 \cap A_2) \Rightarrow b = f(a)$ for some $a \in A_1 \cap A_2$
 $\Rightarrow b = f(a)$ for some $a \in A_1$ and $a \in A_2$
 $\Rightarrow b = f(a)$ for some $a \in A_1$ and
 $b = f(a)$ for some $a \in A_2$
 $\Rightarrow b \in f(A_1)$ and $b \in f(A_2)$
 $\Rightarrow b \in f(A_1) \cap f(A_2)$

$\therefore f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$

* Floor function & Ceiling function

The function $f: \mathbb{R} \rightarrow \mathbb{Z}$ is given by $f(x) = \lfloor x \rfloor =$ the greatest integer less than or equal to x . This function is called greatest integer function or floor function.

eg: $\lfloor 3.8 \rfloor = 3 = \lfloor 3.4 \rfloor = \lfloor 3 \rfloor$
 $\lfloor -3.8 \rfloor = -4$

The function $g: \mathbb{R} \rightarrow \mathbb{Z}$ is given by $g(x) = \lceil x \rceil =$ the least integer greater than or equal to x . This function is called Ceiling function.

eg: $\lceil 3 \rceil = 3$ $\lceil 3.01 \rceil = \lceil 3.7 \rceil = 4$
 $\lceil -3.01 \rceil = \lceil -3.7 \rceil = -3$

* The truncation function is another integer valued function defined on $\mathbb{R}$. This function deletes the fractional part of a real number.

eg: $\text{trunc}(3.73) = 3$
 $\text{trunc}(5) = 5$
 $\text{trunc}(-7.22) = -7$

* Properties of Relations

(i) A relation 'R' on a set A is called reflexive relation if for all $x \in A$, $(x, x) \in R$.

(ii) A relation 'R' on a set A is called irreflexive relation if for any $x \in A$, $(x, x) \notin R$.

(iii) A relation R on a set A is called Symmetric relation if $(x, y) \in R \Rightarrow (y, x) \in R$ for all $x, y \in A$.

(iv) A relation R on a set A is called asymmetric if $(x, y) \in R \Rightarrow (y, x) \notin R$.

(v) A relation R on a set A is called Antisymmetric if $(x, y) \in R$ and $(y, x) \in R \Rightarrow x = y$ for all $x, y \in A$.

(vi) A relation R on a set A is called transitive if for all $x, y, z \in A$, $(x, y), (y, z) \in R \Rightarrow (x, z) \in R$.

* Equivalence Relation

A relation R on a set A is said to be an equivalence relation if R is reflexive, Symmetric and transitive.

* Partially Ordering Relation

A relation R on a set A is said to be a partially ordering relation or partial order if R is reflexive, antisymmetric and transitive.

* Given a finite set A with $|A|=n$, we have $|A \times A| = n^2$.
So there are 2^{n^2} relations on A .

- (i) The number of reflexive relations on A is 2^{n^2-n} .
- (ii) The number of irreflexive relations on A is $2^{\frac{n^2-n}{2}}$.
- (iii) The number of Symmetric relations on A is $2^{\frac{n^2+n}{2}}$.
- (iv) The number of asymmetric relations on A is $3^{\frac{n^2-n}{2}}$.
- (v) The number of antisymmetric relations on A is $2^n 3^{\frac{n^2-n}{2}}$.
- (vi) The number of reflexive and Symmetric relations on A is $2^{\frac{n^2-n}{2}}$.
- (vii) The number of relations which are neither reflexive nor irreflexive are $2^{n^2} - 2 \cdot 2^{n^2-n}$.

1. Determine the number of relations on $A = \{a, b, c, d, e\}$ that are
- (i) reflexive (ii) irreflexive (iii) Symmetric (iv) asymmetric (v) antisymmetric
 - (vi) reflexive & Symmetric (vii) neither reflexive nor irreflexive.

Sol: Here $|A| = 5 = n$.

- (i) No. of reflexive relations = $2^{n^2-n} = 2^{25-5} = 2^{20}$.
- (ii) No. of irreflexive relations = $2^{\frac{n^2-n}{2}} = 2^{10}$.
- (iii) No. of Symmetric relations = $2^{\frac{n^2+n}{2}} = 2^{\frac{25+5}{2}} = 2^{15} = 2^{15}$.
- (iv) No. of asymmetric relations = $3^{\frac{n^2-n}{2}} = 3^{10} = 3^{10}$.

$$(v) \text{ No. of antisymmetric relations} = 2^1 \cdot 3^{\frac{n^2-n}{2}} = 2^5 3^{10}.$$

$$(vi) \text{ No. of reflexive \& symmetric relations} = 2^{\frac{n^2-n}{2}} = 2^{10}.$$

$$(vii) \text{ No. of relations which are neither reflexive nor irreflexive} \\ = 2^{n^2} - 2 \cdot 2^{\frac{n^2-n}{2}} = 2^{25} - 2 \cdot 2^{10} = 2^{25} - 2^{11}.$$

2. Give an example of a relation R , where R is irreflexive, transitive but not Symmetric.

(OR)
Give an example of a relation R on $\mathbb{Z}$, where R is irreflexive, transitive but not Symmetric.

Sol: let us define a relation R on $\mathbb{Z}$

$$\text{i.e. } R = \{(x, y) : x, y \in \mathbb{Z}, x < y\}$$

$$(i) \quad \forall x \in \mathbb{Z}, (x, x) \notin R \text{ because } (x \not< x)$$

$\therefore 'R'$ is irreflexive

$$(ii) \quad \text{For } (x, y), (y, z) \in R \Rightarrow x < y \text{ and } y < z \\ \Rightarrow x < y < z \\ \Rightarrow x < z \\ \Rightarrow (x, z) \in R$$

$\therefore R$ is transitive

$$(iii) \quad \text{For } (x, y) \in R \Rightarrow x < y \Rightarrow y \not< x \Rightarrow (y, x) \notin R \\ \therefore R \text{ is not symmetric.}$$

2. Determine whether the relation R on the set of all lines in $\mathbb{R}^2$ defined by $l_1 R l_2$ if l_1 is perpendicular to l_2 , $l_1, l_2 \in \mathbb{R}^2$, is reflexive, symmetric, antisymmetric, transitive.

Sol: (i) $l_1 \not R l_1 \Rightarrow 'R'$ is not reflexive.

(ii) $l_1 R l_2 \Rightarrow l_2 R l_1$, R is symmetric.

(iii) $l_1 R l_2 \& l_2 R l_1 \Rightarrow l_1 \neq l_2$, R is not anti-symmetric.

(iv) $l_1 R l_2 \& l_2 R l_3 \Rightarrow l_1 \not R l_3$, R is not transitive.